

Dynamic Games

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Dynamic Game

Definition

A Dynamic Game is a game in which players move sequentially over time.

- Dynamic Games can be:
 - finite;
 - infinite;
- Strategies and actions are not equivalent anymore:
 - action- set of choices available to the player when called upon to make a decision;
 - strategies- a complete plan of action for any possible history and point in time;
- Finite dynamic games representation:
 - strategic form;
 - extensive form (game tree);

Example: Roommates Game

- Roommate A chooses whether to play music loud or soft: $\alpha_A^1 = \{L, S\}$;
- Roommate B observes roommate A's choice and chooses between loud and soft as well $\alpha_B^2 = \{L, S\}$;
- The payoffs $u_i(\alpha_A^1, \alpha_B^2)$ for $i = \{A, B\}$ are realized:
 - $u_A(S, S) = 6, u_A(S, L) = 6, u_A(L, S) = 5, u_A(L, L) = 7$;
 - $u_B(S, S) = 3, u_B(S, L) = 4, u_B(L, S) = 4, u_B(L, L) = 5$;

Some Notation

- players: $\{1, 2, \dots, I\}$
- T periods.
- $A_i(h^t) = (a_{i1}(h^t), \dots, a_{in}(h^t))$ - set of actions available to player i given a history h^t .
- $a^t = (a_1^t, a_2^t, \dots, a_I^t)$ - an action profile in period t .
- $h^0 = \emptyset$
- $h^t = (a^0, a^1, \dots, a^{t-1})$
- H^t -set of all h^t .
- The set of all possible actions for player i in period t ,
 $A_i(H^t) = \bigcup_{h^t \in H^t} A_i(h^t)$.
- A strategy for player i in period t : $s_i^t : H^t \rightarrow A_i(H^t)$
- A pure strategy is a sequence $\{s_i^t\}_{t=0}^T$.
- Payoff $u_i : H^T \rightarrow \mathbb{R}$.

Example: Roommate Game

- How many histories are there?
- How many strategies does each player have?

Nash equilibrium

- Let $s_i = \{s_i^t\}_{t=0}^T$.
- $s = (s_1, s_2, \dots, s_I)$ - a strategy profile.
- Then, s constitutes a NE in pure strategies if

$$u_i(s_i, s_{-i}) \geq u_i(s'_i, s_{-i}) \quad \forall s'_i, \forall i$$

Example: Roommate game

- Write the strategic form representation of this game.
- Find all pure strategy NE.

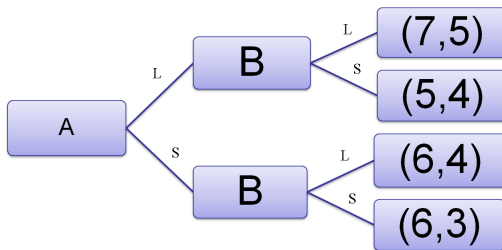
	LL	LS	SL	SS
L	7,5	7,5	5,4	5,4
S	6,4	6,3	6,4	6,3

Nash Equilibrium criticism

- The strategy profile (S, SL) involves a play of dominated strategy by player B.
 - player B plays S as a response to a history $h^1 = L$.
- The only reason this can happen is that this play is off the equilibrium path.
- The threat itself is sufficient to induce a play S by player A.
- Should player A believe such threat?

Extensive Form Representation: Game Tree

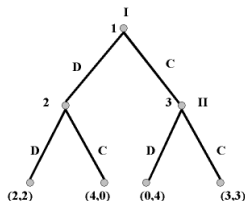
- A game tree is a directed graph that represents
 - the set of players;
 - the order of moves;
 - the available actions for each player;
 - the player's information whenever it is his/her turn to move;
 - the players' payoffs as a function of the moves that were made;
 - the probability distribution over exogenous events;



Game Tree

- A game tree consists of:
 - nodes $x \in X$ with a *partial order* precedence relation $\succ$.
 - $o \in X$ - initial node
 - $z \in Z$ - terminal nodes where z completely determines the path through the tree.
 - $u_i : Z \rightarrow \mathbb{R}$.
 - $i : X \rightarrow I$ - indicates which player moves at each node.
 - $A(x)$ - the set of available actions at node x .
 - $h(x)$ - information set (information available to the player with the move at x).

Information Set



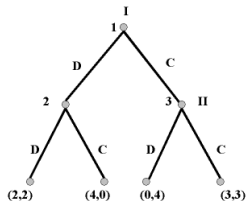
- If $3 \in h(2)$ - then player 2 is uncertain if he is at 2 or 3.
- If $x' \in h(x)$, then $A(x') = A(x)$.
- **Games of Perfect Information:** all information sets are singletons.
 - players move one at a time;
 - each player observes all previous moves;

Strategies Defined over Information Sets

- Let H_i - the set of player i 's information sets.
- $A_i \equiv \bigcup_{h_i \in H_i} A(h_i)$ - the set of all actions for player i .
- $s_i : H_i \rightarrow A_i$.
- $S_i = \times_{h_i \in H_i} A(h_i)$
- $\sigma_i : S_i \rightarrow [0, 1]$ with $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$.
- **Behavior strategy** $b_i(h)$ for player i at information set h is defined as:

$$b_i(h) : A(h) \rightarrow [0, 1] \text{ with } \sum_{a \in A(h)} b_i(a|h) = 1$$

Mixed versus Behavior strategies



Equivalence of Mixed and Behavior Strategies

- perfect recall: no player can forget an information that they once knew.

Kuhn's Theorem

In a game of perfect recall, every mixed strategy is equivalent to the unique behavior strategy that it generates and every behavior strategy is equivalent to every mixed strategy that generates it.

- Without loss of generality, we can restrict attention to behavior strategies (abuse of notation $b_i = \sigma_i$).

Subgame perfection

- An information set h is on the path of the play if it is reached with a positive probability.
- Nash Equilibrium does not put a constraint on the strategies that are not on the path of the play.
- Subgame Perfection: requires that $\sigma_i^*(h)$ is indeed optimal even if $\Pr(h|\sigma) = 0$.

Subgame

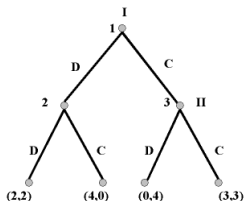
A proper subgame G of an extensive form game T consists of a single node and all its successors with the property that if $x' \in G$ and $x'' \in h(x')$, then $x'' \in G$.

Subgame Perfect Nash Equilibrium (SPNE)

A behavior strategy profile σ is a SPNE if the restriction of σ to G is a NE of G for every G .

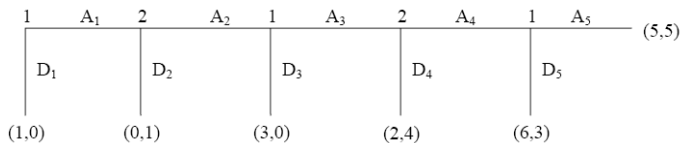
Backward Induction

- Process of identifying the NE of every subgame by moving from the terminal nodes to the initial node and determining the optimal action at each node.



- Backward induction ensures that the resulting NE is subgame-perfect.

Centipede Game



References

- Fudenberg and Tirole- Ch.3